

TOPIC – POLYPROPYLENE APPLICATION EMI SHIELDING

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ABSTRACT

Concerns regarding electromagnetic interference (EMI) and its possible health hazards have grown as a result of the quick spread of wireless communication devices and sophisticated electronic systems, which have increased undesired electromagnetic waves. EMI causes major problems for both commercial and industrial applications by interfering with the operation of electronic components. In this work, the mechanics of electromagnetic interference shielding are discussed, with a focus on the significance of materials that may efficiently attenuate electromagnetic radiation by various internal reflections, absorption, and reflection. Because of their lightweight, flexibility, and ease of processing, conductive polymer nanocomposites—especially those that contain carbon-based materials like carbon nanotubes (CNTs) and carbon black (CB)—have shown great promise as substitutes for conventional metal-based shielding. The developments in polypropylene (PP) composites, which exhibit improved EMI shielding efficacy, are highlighted in this paper.

KEYWORDS

Polypropylene, EMI shielding, Electromagnetic interference, Conductive polymers, Polymer composites, Shielding effectiveness (SE), Carbon nanotubes (CNTs), Conductive fillers.

INTRODUCTION

Unwanted electromagnetic waves are being released at an increasing rate due to the quick development of wireless communication devices and sophisticated electronic systems. In addition to impairing the functionality of nearby electronic components, electromagnetic interference (EMI) poses a health risk to people.[1]

The function of electronic components may be disrupted or deteriorated by electromagnetic interference (EMI), a type of energy that is conveyed by radio or other electromagnetic phenomena.[2]

Problems with electromagnetic interference (EMI) already impair commercial and industrial equipment and have a significant impact on people's daily life.[3]

Electric and magnetic fields operate independently at low frequencies, while electromagnetic radiation (EMR), which produces EMI, is propagated by high frequency waves.[4]

Our environment is overloaded with electromagnetic (EM) radiation as a result of the quick spread of technological devices. Figure 1 illustrates the full spectrum of electromagnetic radiation, which includes radio waves on one side and X-rays and gamma rays on the other. Table 1 lists the applications of these radiations in various communication domains. According to a study published

by the American National Institute of Environmental Health Sciences, male rats' hearts contained obvious remnants of malignant tumours following extensive exposure to radio frequency radiation from cell phones. Other instances of the damaging consequences of this include data theft and electrical gadget malfunctions.[5]

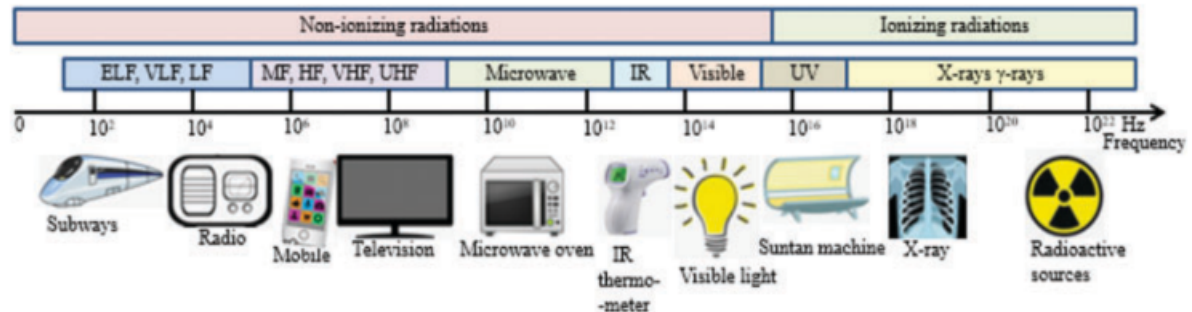


Figure 1. Electromagnetic spectrum, sources and applications.

Figure 1. [5]

Wave category	Frequency range	Band/ITU designation	Application fields
Radio waves	3-30 kHz	Very low frequency(VLF)	Beacon
	30-300 kHz	Low frequency (LF)	Marine communication
	300 kHz – 3MHz	Medium frequency(MF)	AM radio broadcast
	3-30MHz	High frequency	Short wave broadcast, RFID
	30-300 MHz	Very high frequency(VHF)	FM,TV
Microwaves	300 MHz-1GHz	Ultra high frequency (UHF)	TV< microwave oven, mobile phone
	1-2 GHz	L band	Mobile phones, wireless LAN, radar, GPS
	2-4 GHz	S band	Bluetooth
	4-8 GHz	C band	Satellite communication, cordless phone , Wi-Fi
	8-12 GHz	X band	Satellite communication
	12-18 GHz	Ku band	Satellite communication
	18-27 GHz	K band	Satellite communication
	27-40 GHz	Ka band	Satellite communication
	40-75 GHz	V band	Satellite communication
	75-110 GHz	W band	Military research

Table 1. Radio and microwaves in communications [5]

An electromagnetic wave will undergo transmission, absorption, and reflection when it comes into contact with a substance. EMI shielding materials use absorption or reflection to prevent electromagnetic radiation.[6]

Electromagnetic compatibility issues must be resolved in order to avoid the accidental release of electromagnetic radiation (EMR) produced by different electronic devices into the outside world and to eliminate the impact of radiation on specific sensitive components within radio engineering devices.[7] However, the demands of today's increasingly volatile and complicated application environments are beyond the capabilities of conventional EMI shielding materials.[8]

In order to protect modern society's functionality and safety, EMI must be addressed.[9]

Shielding is the only way to prevent such inevitable outcomes because chaos and interference between various electromagnetic radiations have become the main problem of the modern day. EMI shielding is the process of preventing sensitive areas, such as electronic equipment and people, from emitting and/or conducting electromagnetic radiation.[5]

Shielding could be accomplished by absorbing and dissipating radiation power within the material, reflecting the wave, or reducing the amount of signal that passes through the system.[10]

The total of EMI reflection, absorption, and multiply reflection can be interpreted as the electromagnetic interference shielding effectiveness. Electric conductivity is the fundamental requirement for reflection shielding, whereas materials with electric or magnetic dipoles make good absorption shielding possibilities.[11]

Shielding effectiveness (SE) for electromagnetic interference (EMI) refers to defence against radio waves (Rws) or microwave radiation (MW).[12]

Decibel (dB) measurements are frequently used to quantify the shielding effectiveness (SE) of electromagnetic wavelengths (EMWs).[13]

The development of conductive polymer nanocomposite EMI shielding materials is therefore fuelled by the need for materials with a high EMI shielding effectiveness (EMI SE) to address these problems.[1]

By reflecting the field waves backward or by absorbing and dissipating the radiation within the material, electromagnetic shielding materials can block and reduce the signal that is emitted.[14]

The ability of a material to attenuate incident radiation through mechanisms of reflection and absorption determines its electromagnetic interference shielding effectiveness (EMI SE). The first occurrence, which is significant for electrically conducting materials, is facilitated by the presence of mobile charge carriers that interact with the electromagnetic waves. The process by which electromagnetic radiation is converted into heat or another form of energy and/or dissipated is the basis for the absorption mechanism. Electric and/or magnetic dipoles that can interact with electromagnetic radiation are used to produce this property. In addition to being crucial for stealth technology and defence to lessen the radar detectability of key targets, materials whose primary feature is an absorption mechanism are gaining attention since they may prevent self-emission of electromagnetic waves.[15]

Controlling the transmission and pollution of electromagnetic waves requires the use of materials that are EM shielding, wave-absorbing, and wave-transparent.[16]

When compared to conventional EMI shielding materials, multi-layered materials can enhance reflective surfaces, allow for flexible control over the composition of each layer, and enhance EMI shielding performance. Fu et al. created sandwich-shaped foamed polypropylene/CNT nanocomposites with a micro-foamed core layer and nano-foamed skin layers, which improved the EMI shielding capabilities. By using supercritical CO₂ foaming, Wang et al. created a sandwich-shaped, lightweight, and flexible thermoplastic polyurethane/CNT foam. The structure's frequency-selective EMI shielding ability was demonstrated by the results.[17]

The increased awareness of health regimens has led to a growing variety of items with electromagnetic shielding effectiveness (EMSE) and UV-blocking capabilities, most frequently in the shape of staple fabrics. Carbon-based materials, conductive polymers, and metals are widely employed as electromagnetic wave shields.[18]

Because of their strong electrical conductivity and general shielding efficacy, metals have been utilised extensively during the past ten years to address the problem of electromagnetic shielding interference. Nevertheless, metals have a number of drawbacks, including high mass density, corrosion, and processing difficulties.[19]

Metals have long been preferred for shielding against electromagnetic interference (EMI). However, compared to polymers, their stiffness, ease of machining, corrosion susceptibility, and significant weight limit their employment in aeroplanes and contemporary smart devices. Conversely, several ceramic materials, such as SiC and Si₃N₄, have demonstrated promise as effective alternatives.[9]

Although metal fibres are frequently utilised to create electromagnetic shielding materials with superior electromagnetic shielding properties, metal-based materials also have several significant drawbacks, including high weight, challenging manufacturing, and poor chemical stability.[20]

Comparatively speaking, polymer materials have several benefits over metals, including low cost, light weight, ease of processing, strong corrosion resistance, and so forth. As a result, they have found extensive use in a variety of everyday and industrial contexts. Because of their many benefits, carbon-based nanomaterials are frequently used to create composite materials made of electromagnetic shielding polymers.[21]

Instead of reflection, as happens in metals, the polymers primarily permit EMI waves to penetrate through the surface and cause absorption phenomena.[22]

Although polymers often have modest mechanical properties, their flexibility allows them to be used in situations where mechanical properties are required. Adding inorganic elements to polymers improves their mechanical properties, such as stiffness, tensile strength, and modulus.[23]

Additionally, these materials must demonstrate multidimensional properties including hydrophobic performance, thermal insulation, and thermal conductivity in order to be in line with a variety of application scenarios.[24]

It is well known that electrical conductivity is crucial and directly affects how efficiently a polymer nanocomposite shield blocks electromagnetic interference.[25]

By reflecting and absorbing electromagnetic waves, electronic conductive materials can act as a shield to reduce or even completely eradicate these interference problems.[26]

Numerous polymer composites have been investigated for EMI shielding over time.[27]

One of the most significant polymers for which enhanced electrical characteristics and EMI/RFI shielding capabilities are occasionally sought is polypropylene. One of the most used polymers, polypropylene is ideal for injection moulding and is almost indefinitely recyclable.[28]

Because of its favourable mechanical qualities, chemical resistance, lightweight, recyclability, and superior processability, PP is a semi-crystalline thermoplastic that finds extensive use. The EMI shielding effectiveness (SE) of these composites was higher than that of those made using a melt-blending technique.[29]

Because PP has a low melting point, it is beneficial to connect strongly with nylon mesh in order to ensure that the composites have good mechanical qualities.[30]

In the S-band area, the addition of 20% conductive fibres with a thickness of 2 mm raises the EMI to 40 dB. Pure PP has no EMI shielding properties. In the X-band area, a 0.7 mm thick polypropylene (PP)/graphene composite exhibited an EMI-SE of 29.3 dB with an EC of 0.0409 S cm⁻¹.

Due to the conductivity produced by the added fillers, the SER is primarily responsible for the EMI-SE capability of CB-PP-PS-SBS.[31]

Propylene monomers are united in an unbent chain to form PP's unique molecular structure, which accounts for its adaptability. Increased crystallinity, better chemical resistance, lower density, and adequate mechanical strength are some advantages of this linear structure [4]. Because of these

characteristics, PP can be used in a wide range of items, including medical equipment, automotive components, and packaging materials. Mechanical strength and electromagnetic shielding are enhanced in PP composites supplemented with jute fibres or carbon nanotubes (CNT).[32]

Because of its exceptional mechanical qualities, ease of manufacturing, and recyclability, polypropylene (PP) and its composites are the most widely used plastics in automobiles. A significant amount of the automotive industry is composed of polymer composites because of its affordability and low weight. When recycling at high temperatures and high shear pressures, macromolecular chain scissions cause the molecular weight to decrease, which results in poor mechanical qualities for recycled materials. It is possible to add different fillers to the waste to enhance its mechanical performance.[33]

Considering that PP is one of the most widely used polymers, the best for injection moulding, and practically infinitely recyclable, it is crucial to develop PP-based composite materials with additional value and specific functionality (such as electrical conductivity and EMI shielding). The production of solid-state composite materials that may be utilised as EMI shielding casings for electronics and automotive applications is described in this paper.[28]

In the conductive and EMI shielding sectors, polypropylene (PP), a thermoplastic, is favoured over other polymeric material matrices because of its notable benefits, which include high recyclability, chemical resilience, and processing ease.[34]

The sheath is made of high-strength, UV-resistant 150 D polypropylene (PP) filaments, while the core is made of 0.05 mm and 0.08 mm diameter stainless steel (SS) filaments.[35]

These polymers may be melt mixed with traditional thermoplastics like polyethylene (PE), polypropylene (PP), or polystyrene (PS) due to their great thermal stability and acceptable electrical conductivity values.[36]

Apart from polymers, several nanostructures have also been studied in conjunction with polymers, including carbon and inorganic nanoparticles.[37]

Composite materials composed of conductive and magnetic additives have drawn a lot of interest lately. Because of their extremely high electrical conductivity and large number of surface functional groups, MXenes have demonstrated significant EMI shielding potential. Inspired by the structure of coral reefs, Sun et al. used MXene, carbon nanotubes (CNTs), and Fe₃O₄ to create a high-performance polyurethane composite elastomer.

Because of its precisely calibrated multidimensional coral reef-like layered heterostructure, the composite elastomer can function efficiently in the -20 to 100 °C range and has a minimum reflection loss (RL_{min}) of -54.81 dB. Mao and associates.[38]

Carbon nanomaterial-based polymer composites combine the optical and electrical qualities of highly electrically conductive and absorptive carbon fillers with the mechanical strength and flexibility of the polymer matrix.[39]

Such materials must be extremely effective, lightweight, and long-lasting in order to provide effective EMI protection. Composites of carbon-based polymers are among these materials.[40]

When considering mechanics, thermodynamics, and electrical, carbon materials that are frequently employed as reinforcing components—such as graphene, carbon fibres, and carbon nanotubes—have more usefulness.[41]

Because of their inherent high electrical conductivity and high electromagnetic (EM) wave reflection, carbon-based materials and metals are two of the most commonly utilised materials for EMI shielding. Common polymers are typically ineffective or less effective due to their non-conductive nature, unless they are paired with additional functional materials and/or components.[42]

Conductive polymer composites (CPCs) have garnered a lot more attention lately than standard metallic shields because of their benefits, which include low cost, easy processability, lightweight, and corrosion resistance. These served as the basis for the initial batch foaming of polypropylene (PP)

composite beads containing 5–30 weight percent CB particles using subcritical CO₂. These beads were then utilised as building blocks for the steam-chest moulding process to create large-scale foams. In this case, CB was used as an inexpensive dielectric loss material that was anticipated to work with the foam structures to give the composites better EMI SE and high absorption loss to electromagnetic waves.[43]

Because of their exceptional processability, corrosion resistance, and lightweight nature, conductive polymer composites (CPCs) hold great promise as EMI shielding materials. In order to facilitate the injection moulding of segregated CNT/isotactic PP (iPP) composites, low-viscosity atactic polypropylene (aPP) was recently introduced as a flow accelerator.[44]

Thermoplastics like polypropylene (PP) and polyethylene, as well as thermosetting resins like epoxy resin, are frequently utilised in CPCs matrix for EMI shielding materials.[45]

Both small and large-scale industrial uses, including electrical conductivity, EMI shielding, charge storage, and other mechanical applications like sporting goods, can benefit from the use of polymer blends containing carbon nanotubes.[46]

Because of their own extraordinary electrical conductivity (10⁶–10⁷ S/m), outstanding thermal conductivity (3500 W/Mk), high UV adsorption, high tensile strength (30–45 GPa), yield strain (5.3–5.8%), light weight, low density, and flexibility, carbon nanotubes (CNTs) appear to be the most suitable in this regard.[47]

One of the promising materials for attenuating electromagnetic interference (EMI) is carbon nanotubes (CNTs)/polymer nanocomposites. CNT/polymer composites are straightforward to produce, lightweight, flexible to application requirements, and versatile. The dispersion and distribution of CNTs are important elements that determine how much CNT is required to achieve a given level of EMI attenuation. The conditions and method of composite processing affect the dispersion and distribution of CNTs.[48]

Since EMI shielding properties require a conductive filler, carbon nanotubes (CNTs) with excellent electrical conductivity (10⁴–10⁶ S · cm⁻¹), thermal properties (3000–6000 W · m⁻¹K), ultra-high strength (10–100 GPa), and high modulus (~1.0 TPa) have been demonstrated to be a highly effective conductive nanofiller for this purpose [23–25]. For example, by adding 4.6 vol% CNT to PP random copolymer, Verma et al. showed an improvement in DC conductivity, tensile strength, and a high EMI SE value of 47 dB. A hybrid filler system (1.75/1.75 vol% CNT/stainless steel fibre) in PP can obtain a high EMI SE of roughly 47.5 dB, according to another study.[33]

By maximising the synergistic effects of the ferroelectric phase (BT) and MWCNT fillers, Jin et al. produced CPCs with ferroelectric barium titanate (BT) and multiwalled carbon nanotube (MWCNT) fillers, achieving dependable dielectric characteristics.[49]

THEORY

INTRODUCTION TO EMI SHIELDING AND ITS IMPORTANCE

Electromagnetic interference (EMI) is the interference or disruption of an electromagnetic signal generated or received by another electrical device. Depending on the source and transmission path, interference might be conducted or radiated. One example of radiative interference is the radiation produced by a device that travels through the air away from the apparatus. In the electromagnetic spectrum, electromagnetic interference (EMI) happens between 10⁴ and 10¹² Hz.[50] A type of environmental contamination known as electromagnetic interference (EMI) is brought on by both electronic equipment and natural occurrences like solar flares and electrostatic discharge. Depending on the technology used in their creation, the extensive usage of smart electrical and communication devices has led to an increase in electromagnetic radiation emission across a broad frequency range. Long-term exposure to this radiation may pose health risks to people and have an impact on a number of areas of daily life, such as personal mobile phones, communications, and bank security. It can also alter the sensitivity and functionality of other electronic devices. The creation of materials that can provide radiation protection is therefore a matter of increasing importance.[15] Electromagnetic noise (EMN), also known as un-compatible environments (UCE), is

created by the growth of digital networks and sensitive remote-controlled systems that require high-quality, densely built electronic control systems. Electrodynamic and basic field currents are also important sources of EMI. [4]

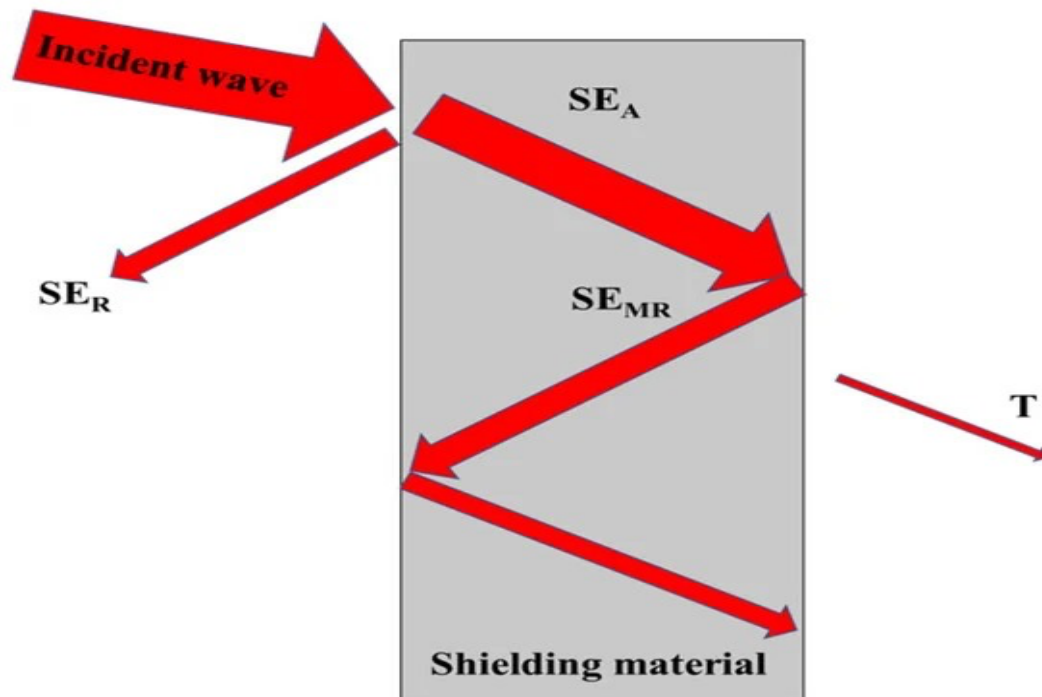


Figure 2. Transformation of the electromagnetic radiation (EMR) at the shielding material[4]

While electromagnetic radiation (EMR), which causes electromagnetic interference (EMI), is propagated by high frequency waves, electric and magnetic fields act independently at low frequencies. Radio and television transmissions, radar, aircraft, electromagnetic missiles, the battlefield, Bluetooth, wireless local area networks (WLAN), and remote controls are the sources of electromagnetic interference (EMI), which is caused by radiative coupling.[4] Therefore, by putting a shielding material between the device and the source, EMI can be avoided. SE, which is measured in decibels, is the ratio of the transmitted power to the incident radiation power.[50]

EMI SHIELDING MECHANISM

Reducing EMI emissions from the source, altering or changing the electromagnetic energy's propagation path, and enhancing the susceptor's immunity are the usual methods for EMI management. The phenomena of reflection, absorption, and transmission occur once electromagnetic waves impinge on a surface. Depending on the properties of the material and the type of waves, these waves are attenuated by reflection, multiple reflection, and absorption, a Furthermore, the attenuation of electromagnetic radiation is also influenced by multiple reflections.[5]

The main mechanisms of electromagnetic interference shielding include material absorption, reflection, and numerous internal reflections of electromagnetic waves. Reducing or eliminating electromagnetic wave interference with systems, equipment, and the environment is its main goal. For materials that have High conductivity, a suitable relative dielectric constant, magnetic permeability, and a large interior surface area are necessary for superior shielding efficacy. The interaction between incident electromagnetic waves and free charge carriers on the material surface results in reflection, which is regarded as the main shielding mechanism for electromagnetic

interference. Shielding materials must have free charge carriers, like electrons and/or holes, and good conductivity to allow these carriers to pass through and improve EMI shielding in order to produce effective reflection.[51]

Nonetheless, possessing high conductivity alone is not both a necessary and sufficient condition since effective connectivity plays a more crucial role in the conduction pathway. As a secondary mechanism for EMI shielding, absorption is improved by materials that demonstrate dielectric or magnetic losses. For absorption loss to occur, it is crucial that the shielding material contains magnetic dipoles and/or electric dipoles.[51]

Materials with elevated dielectric constants are capable of producing electric dipoles, while those with high magnetic permeability can create magnetic dipoles. The mechanism of multiple internal reflections, another aspect of EMI shielding, involves reflections that take place on different surfaces or interfaces within the shielding materials. This mechanism necessitates materials with extensive surface areas and interface areas, such as porous, hollow, multi-layered, and foam types. However, this effect is primarily observable in thin films and porous configurations.[51]

An electromagnetic wave can be transmitted, reflected, or absorbed by a material when it strikes its surface. The electromagnetic interference shielding effectiveness of a substance determines its capacity to lower EM waves. The ratio of incident energy (E_i), represented by the incident EM wave to transmitted energy (E_t), represented by the transmitted EM wave yields the total EMI SE (SET), expressed in decibels (dB): $= -20\log(E_t/E_i)$ SE(dB)[15]

The reduction in incident power/field magnitude upon transmission across the shield is the standard metric used to quantify EMI shielding. The sum of the effects of a material's two primary mechanisms—reflection SE (SER) and absorption SE (SEA)—is the total EMI SE (SET). Reflection mostly depends on the material's electrical conductivity, whereas absorption is linked to dielectric/magnetic polarization or energy dissipation. The reduction in incident power/field magnitude upon transmission across the shield is the standard metric used to quantify EMI shielding. The sum of the effects of a material's two primary mechanisms—reflection SE (SER) and absorption SE (SEA)—is the total EMI SE (SET). Absorption relates to dielectric or magnetic polarization, as well as energy loss, whereas reflection is primarily influenced by the material's electrical conductivity. CNTs are generated through the foaming process in CNT-TSM-PDMS composite foams.[29]

A material's capacity to attenuate incident radiation through mechanisms of reflection and absorption determines its electromagnetic interference shielding effectiveness (EMI SE). The first occurrence, which is significant for electrically conducting materials, is encouraged by the presence of mobile charge carriers that interact with the electromagnetic waves. The method of absorption relies on the dispersion of the radiation and/or the transformation of electromagnetic energy (EM) into heat or another form of energy. Electric and/or magnetic dipoles that can interact with electromagnetic radiation are used to produce this property. In addition to being crucial for stealth technology and defense to lessen the radar detectability of strategic targets, materials whose protection primarily consists of an absorption mechanism are gaining attention because they can prevent self-emission of electromagnetic waves.[15]

The miniaturization of electronic and telecommunication devices necessitates the use of flexible and processable materials that can be easily molded into various sizes and shapes. Consequently, polymer composites infused with specific conductive and/or magnetic fillers (commonly referred to as extrinsically conducting polymers) have been thoroughly investigated for applications in electromagnetic interference (EMI) shielding and microwave absorption. Numerous significant studies and reviews concentrate on carbon-based nanoparticles (such as carbon nanotubes (CNTs), expanded graphite (EG), graphene nanoplatelets (GNPs), and carbon black (CB)) as fillers because of their excellent conductivity, mechanical strength, and resistance to corrosion.[15]

The effectiveness of electromagnetic interference shielding can be perceived as the collective result of reflection, absorption, and multiple reflections of EMI. The fundamental requirement for reflection shielding is the electrical conductivity of the material, while materials that have magnetic or electric dipoles are ideal candidates for absorption shielding. Multiple reflections occur through internal

reflections of electromagnetic wave radiation from various inhomogeneities and phase interfaces present within the shield.[15]

EMI SHIELDING MATERIALS

Many shielding materials with good shielding qualities have been produced up to this point . Regretfully, these materials rely too heavily on increasing electromagnetic (EM) wave reflection. These waves' reflection maintains electromagnetic radiation in the atmosphere, which can impact other devices or reflect into the environment, leading to more severe EMI pollution issues , also known as EMI smog.[33]

A good shielding material should exhibit at least the minimum SE that at the moment is 20 dB, which corresponds to shielding 99% of incident EM radiation , favorably in a broad frequency range.[39]

By either absorbing and dissipating the radiation within the material or by reflecting the field waves backward, EM shielding materials can block and reduce the signal that is emitted . Conventional metal shields can successfully reduce electromagnetic interference (EMI) produced by 5G applications, but they often have some disadvantages, like limited corrosion resistance, bulkiness, weight, and volume. When compared to metal-based materials, conductive polymer composites are a good substitute because of their low density, flexibility, strong chemical stability, and ease of manufacturing. Because of this, polymer-based materials are becoming more and more relevant. They perform well when filled with fillers that improve their characteristics, including carbon nanotubes (CNTs), which removes the drawbacks of metal-based materials.[14]

Among typically used materials for EMI shielding, carbon-based materials and metals are two of the most widely used, owing to their intrinsically high electrical conductivity and high electromagnetic (EM) wave reflection[42]

Metals are excellent conductors of electricity and may reflect EM waves; hence, metals are widely used in EMI shielding applications.[22]

Because of their high conductivity and capacity to block electromagnetic fields, certain metals—such as copper, aluminium, and stainless steel—are frequently employed in EMI shielding; nevertheless, their weight, stiffness, and resistance to corrosion make them difficult to use.[13]

However, the shielding mechanism in metals is dominated by the reaection of EM waves, which is not always a desirable option.[22]

Metals are typically employed to block incident electromagnetic waves, however they present problems such low processability, corrosion, and heavy weight. Secondary electromagnetic pollution may result from metals' shallow skin depth, which causes the majority of the radiation power to be reflected rather than diminished.[29]

Because of their anti-corrosion, radiation-protective, and lightweight qualities, polymers have been used in this situation. The effectiveness of traditional electromagnetic shields to block these radiations is limited. Several thermosets and thermoplastics, such as polyethylene, polypropylene, polystyrene, phenolics, epoxy, and others, have been investigated for EMI shielding.Incoming electromagnetic waves can be effectively shielded by conducting polymers.[37]

Polymer matrices commonly used for EMI shielding materials are both thermoplastics like PC, PP, PE, PS, PLA, PMMA , ABS or PVDF and thermosets like epoxy resins, PDMS or PU.[26]

With specialized uses in energy storage, sensors, anticorrosive materials, electrostatic charge dissipation (ESD), electromagnetic interference shielding (EMI), organic light emitting diodes (OLEDs), solar cells, and catalysis, conducting polymers have become a significant class of materials.[36]

Polypropylene (PP) is the most effective self-reinforced polymer because to its superior impact performance and processability. The tensile and toughness of self-reinforced polypropylene

composites are superior to those of isotropic polypropylene and conventional fiber-reinforced composites, respectively.[52]

Another thermoplastic polymer that is frequently used to create flexible and durable olefinic foams with a variety of uses in the automotive, roofing, upholstery, biomedical, and construction industries is polypropylene. Its use as an effective high-performance microwave absorber is, nevertheless, somewhat restricted and has not yet been identified. Ju and colleagues created lightweight polypropylene (PP)/carbon nanotubes (CNTs)/carbon black (CB) nanocomposite foams by using a similar approach taken by Chen et al. Here, PP powders and CNT/CB powder were combined in varying weight ratios using a grinder set to 20,000 rpm/min and 100 °C. The hybrid combination of the fillers was probably the cause of these porous foams' exceptionally low density (0.082–0.101 g/cm³) and exceedingly low percolation threshold (0.016 vol.%). A peak EMI SE value of 20 dB was achieved at a thickness of 2.5 mm, with a particular EMI SE reaching as high as 72.23 dB·cm³/g, exhibiting a behavior predominantly characterized by absorption-based shielding.[25]

Because it is lightweight, flexible, and easily processed, polypropylene can be used in a variety of ways. Because polypropylene is a polymer with low intrinsic conductivity, conductive fillers must be added to increase the efficiency of EMI shielding.[31]

Pure PP, mPP, and its composites were examined for morphology using a JSM-IT300 scanning electron microscope (SEM). Pure PP and mPP water contact angle (WCA) values were measured using a JY-82B Kruss DSA equipment. A Nicolet 6700 spectrometer was used to record the Fourier transform infrared (FTIR) spectra of PP and mPP in the 4000–400 cm⁻¹ spectral range. The composites' electrical conductivity was measured using a TH2690 high resistance meter. With a PNA N5224B network analyzer, the composites' EMI shielding effectiveness (SE) was assessed in the X-band to determine the S parameters.[45]

Polypropylene (PP) filaments are 150 D fine and offer a UV-blocking effect. Following that, the woven textiles are laminated using various lamination-layer numbers and lamination-layer angles for testing of ultraviolet radiation and electromagnetic shielding efficacy.[35]

PP is a thermoplastic that is crystalline, strong, and stiff. Throughout its history, PP has seen numerous advancements and widespread acceptance, becoming a standard in many industries thanks to its affordability, versatility, and superior performance in a variety of applications. Notwithstanding its many advantages, PP composites based on natural and synthetic polymers have certain drawbacks, such as lengthy preparation times, high manufacturing costs, limited demand, susceptibility to moisture contact, and many more. Its application is rapidly growing across a number of industries, such as consumer goods, packaging, textiles, and automobiles. Over the past few decades, researchers have focused more on enhancing the performance of PP through blending or combining with other nanomaterials. PP is being used in new domains and has a variety of applications in nanotechnology.[32]

PP is one of the thermoplastic resins widely applied in the industrial field, which has a regular crystal structure, easy processability, good impact resistance, and good flexural property.[41]

For broad applications, the PP's specific electromagnetic shielding capability is especially crucial. Therefore, the PP and PP-MA18's electromagnetic shielding capabilities were examined. [53]

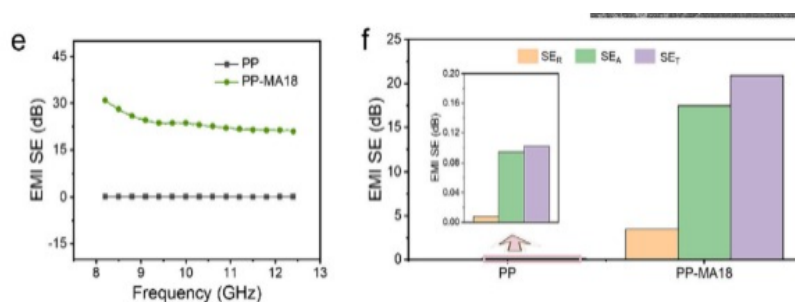


Figure3 . Impact of X-band EM shielding effectiveness of PP and PP_MA18; comparison of SE~total SE~A and Se~B of PP and PP-MA18 at 12.4 GHz[53]

The doping of MX@LNP greatly increased the electromagnetic shielding effectiveness (EMI SE) of PP-MA18. When it came to incident electromagnetic waves, pure PP was almost transparent. In the X-band, however, PP-MA18 had an average EMI SE of 20.915 dB. For commercial applications, this value essentially satisfied the minimal requirements. The total electromagnetic shielding effectiveness (SETotal), absorption shielding effectiveness (SEA), and reflection shielding effectiveness (SER) for PP and PP-MA18 from 8.2 to 12.4 GHz were compared in order to examine the electromagnetic shielding mechanism of the produced PP-MA composites.

it is clear that the SETotal, SEA, and SER values for PP-MA18 were 20.915, 17.441, and 3.474 dB, respectively, whereas the corresponding values for PP were almost 0 dB. The main mechanism contributing to the shielding of PP-MA18 was primarily due to the loss associated with the absorption of electromagnetic waves.[53]

Polypropylene (PP) is a crucial material that has found extensive use in everyday life and various industries . However, it is highly flammable and can decompose rapidly when exposed to fire, producing a significant amount of heat as well as toxic and hazardous gases due to its linear arrangement of carbon and hydrogen, which poses risks to public and societal safety . Consequently, there is a pressing need for high-quality flame-retardant PP. Moreover, considerations for environmental protection are increasingly important in ensuring safety in everyday life.[53]

Propylene monomers are united in an unbent chain to form PP's unique molecular structure, which accounts for its adaptability. Numerous advantages are suggested by this linear configuration, such as improved mechanical strength, higher chemical resistance, decreased density, and enhanced crystallinity. Because of these characteristics, PP can be used in a wide range of items, including medical devices, automotive components, and packaging materials. [32]

The use of PP in the composite as is performing a significant role in the thermoplastic type composite. Diferent man-made, natural and high-performance fber has been used as reinforcement over the century.[32]

Techniques for Making Composites Based on Polymers
By creating a composite using the hydrothermal approach, the highest total shielding efficacy in the Ku-band was attained up to 60 dB with a reflection loss of -61.4 dB. The maximum total shielding efficacy in the K-band frequency spectrum was 51.4 dB, 51.1 dB, and 51 dB employing the solution processing method, Hummer's approach, and ultrasonication methodology. On the other hand, the solution processing method produced a reflection loss of -61 dB. Chemical vapor deposition produced the highest total shielding effectiveness of 77 dB in the Ka-band frequency range.[19]

ELECTROMAGNETIC SHIELDING PROPERTIES

The mechanical properties of polymer composite materials can be mostly controlled by the choice of components, the percentage and type of addition in the matrix (binder), the geometry and orientation of the fillers.[54]

Because of its many benefits, such as high specific strength, ease of manufacturing, and versatile performance, polymer-based EMI shielding composites have found extensive use in microelectronics technology, aircraft, and other domains. The benefits of EMI shielding fabrics made of MXenes or polymers are their softness, breathability, low weight, ease of manufacturing, and robust designability.[24]

The EMI shielding industry is one use for polymer composites with high electrical conductivity values. Generally speaking, as the conductivity of the composite increases, so does the EMI attenuation. [48]

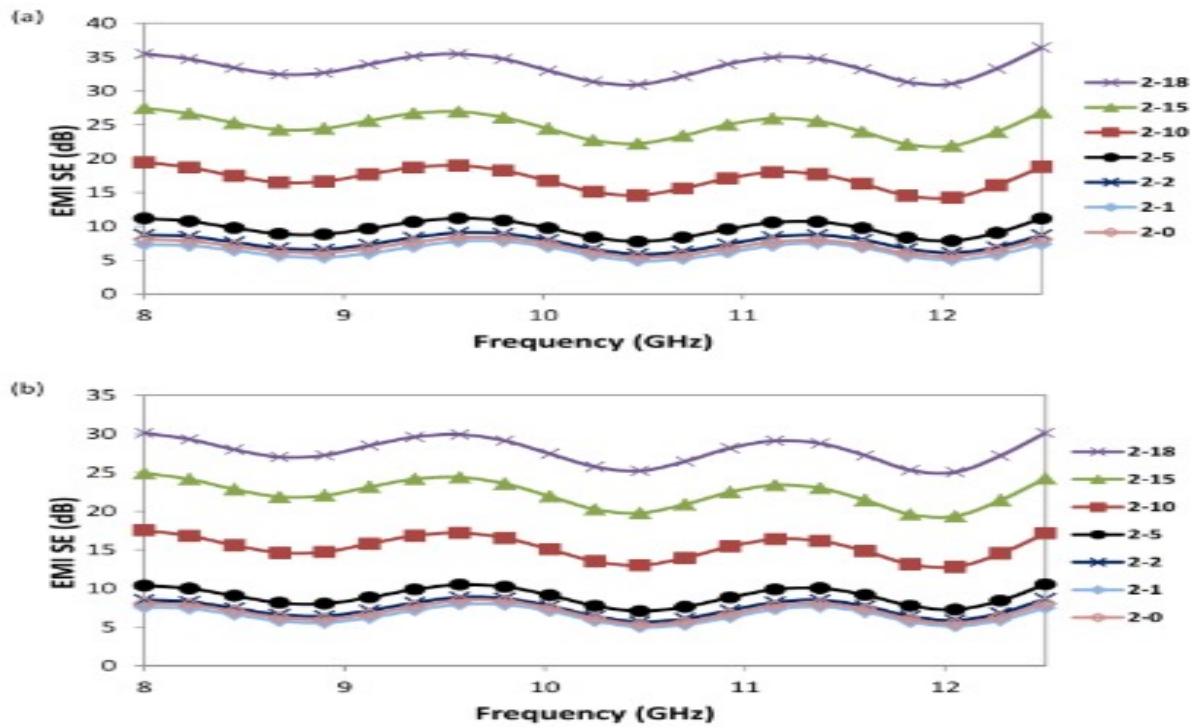


Figure 4. EMI SE of CNT :CB filled PP/PE blends as function of CB+NT:CB mass fraction and frequency (a) 90/10 PP/PE blends (b)50/50 PP/PE blends. [48]

The EMI SE of 2 pph.% CNT-filled PP/PE blends in the X-band frequency range is shown in Fig. 6 as a function of CB concentration and PP/PE volume ratio. It is evident for both blends that as the CB mass fraction rose, so did the blend's EMI SE. For mixes with an average EMI above 20 dB, the X-band EMI SE showed a wavy profile with a difference of about 5 dB between the lowest and highest SE. The variation between the highest and lowest SE within blends with an average EMI of less than 20 dB[48]

For blends that have an average EMI below 20 dB, the variation between the highest and lowest SE in the X-band range is approximately 3 dB. Figure 7 illustrates the overall EMI SE of the 2 pph.% CNT-filled PP/PE blends at 8.0 GHz, plotted against the PP/PE volume ratio and CB content.[48]

Electrical conductivity, conductive network development, and conductive filler concentration all affect how well a polymer nanocomposite shields against electromagnetic interference. The effectiveness of CN in EMI shielding and EM wave absorption was confirmed. As a crucial component [55]

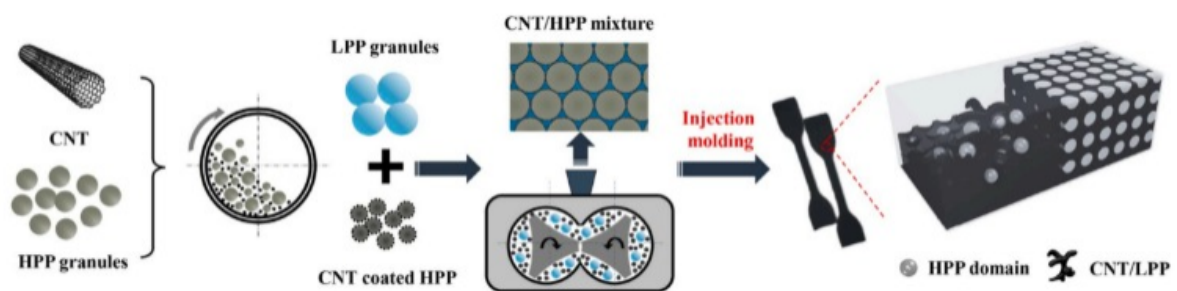


Figure 5. Schematic for the preparation of the segregated CNT/PP composites. [44]

First, the electrical conductivity of the CNT/PP composites is examined in relation to the parameters of the EMI shielding material; the findings are shown in Fig. 5. The enhanced conductive network is responsible for IS-x's increased electrical conductivity as its CNT content rises. The electrical conductivity of IS-2.0 is already 3.3 S/m, meeting the requirements of the commercial EMI shielding application. At 14.4 S/m, the maximum electrical conductivity, IS-3.5 is 2057 and 1.5 times more than IR-3.5's 0.007 S/m and CM-3.5's 9.6 S/m.[44]

The attainment of enhanced electrical conductivity in IS-3.5 can be attributed to its favorable segregated structure that optimizes CNT usage. In IS-3.5, the development of the segregated network demonstrates that a considerable quantity of CNTs is concentrated at the interfaces of HPP domains and heavily overlapped, which facilitates electron transmission. Conversely, CM-3.5 exhibits more defects and barriers (such as voids and the polymer matrix), resulting in diminished electrical conductivity.[44]

Melt blending was used to create PP/CNT polymer composites. A Novocontrol Alpha Broadband Dielectric/Impedance Spectrometer was used to measure the electrical conductivity, and a Fluke 289 multimeter was used to confirm the results. Anritsu Shockline 2 ports VNA (MS 46522B) and two different kinds of measuring cells were used to assess the EMI shielding of PP/CNT composites. A coaxial cell (EpsiMu Multiwave PE 13 mm) has been utilized to assess the EMI shielding efficacy (EMISE) between 50 kHz and 8 GHz.[26]

Polymer blends filled with CNTs can be utilized in both small and large-scale industrial applications, such as electrical conductivity, EMI shielding, charge storage and other mechanical applications like sporting goods.[46]

PP—CNT composites' efficacy in EMI shielding in coaxial cells This composite's shielding efficacy is 42 dB at low frequencies (between 50 MHz and 1 GHz) and rapidly rises to 60 dB at 8 GHz in the high frequency region. The skin depth of this composite material (10 vol% CNT) is 0.23 mm at this frequency of 8 GHz, making it thinner than the sample, which has a thickness of 1 mm. Effectiveness of PP—CNT composites in waveguide cells for EMI shielding Even with this relatively low CNT concentration, a film with a thickness of 0.4 mm shows a non-negligible shielding of 20 dB, whereas a sample with a thickness of 2.8 mm achieves shielding of over 70 dB at 18 GHz.[26]

Because of their exceptional performance and numerous prospective uses, carbon nanotubes, or CNTs, are garnering ongoing attention on a global scale. Research employing waste plastics as raw materials has been spurred by the potential benefit of using them to replace alternative carbon sources in the petrochemical industry for the production of carbon nanotubes. Tang et al. created MWCNTs by catalyzing the combustion of PP by combining PP with maleized polypropylene (PP-MA), nickel catalyst (Ni-Cat), and organically modified clay (OMC). High-quality and carefully cultivated carbon nanotubes can be generated with the help of particular catalysts for the carbonization process.[51]

A kind of nanomaterials known as carbon nanotubes (CNTs) are made of rolled-up graphene sheets with sp² bonds connecting carbon molecules. It's interesting to note that depending on their sidewalls and atomic structural arrangement, these carbon nanotubes can be single-walled (SW) or multi-walled (MW).[27]

The first test of PP-CNT composites' EMI shielding efficacy was conducted in a coaxial cell between 50 MHz and 8 GHz. Applications using this broad frequency range include FM radio, Wi-Fi, Bluetooth, cellphones, and GPS waves. The sample is subjected to the incidence of a progressive plane electromagnetic wave in the far field regime when using this kind of cell. Fig. 5-a shows examples of SE vs frequency for CNT composites with a thickness of 1 mm and 5 and 10 vol%. According to the theoretical model, the shielding rises above a critical frequency (where both reflection and absorption contributions are present) and remains constant at low frequencies (defined by the reflection regime).[26]

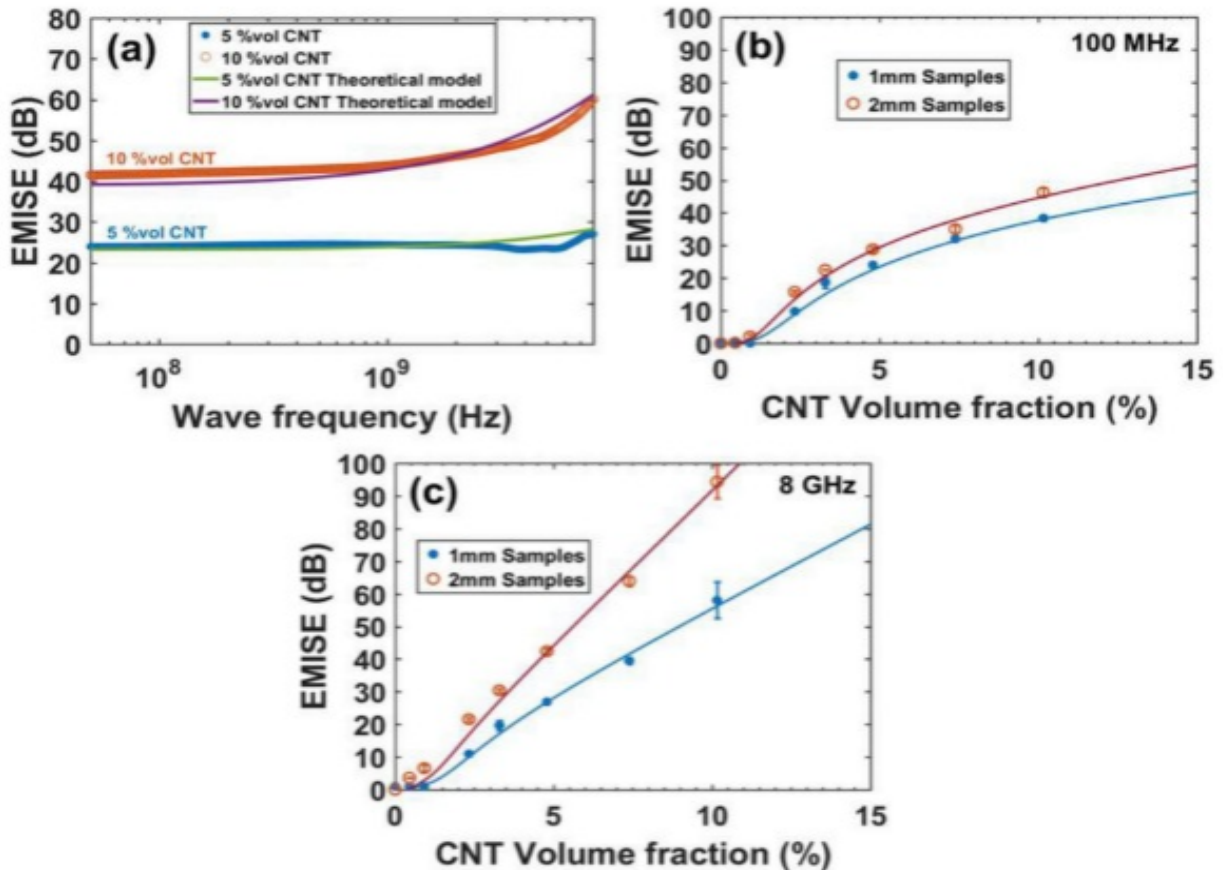


Figure 6 . EMI shielding effectiveness of PP/CNT composites measured in the coaxial cell from 50 kHz to 8 GHz. [26]

(a): Experimental shielding of 5% and 10% vol. CNT composites versus incident wave frequency (thick lines). The experimentally measured conductivity of these composites is used in the theoretical model to get the theoretical shielding curves (thin lines).

(b): Experimental shielding of composites versus CNT volume fraction (dots) at 100 MHz incident wave frequency. By taking into account the experimental percolation law into the shielding theoretical model, the theoretical shielding curves are plotted as thin lines.

(c): Same as before but at 8 GHz incident wave frequency.

With the 10 vol% CNT composite, this behavior is particularly apparent. This composite's shielding efficacy is 42 dB at low frequencies (between 50 MHz and 1 GHz) and rapidly rises to 60 dB at 8 GHz in the high frequency region. The skin depth of this composite material (10 vol% CNT) is 0.23 mm at this frequency of 8 GHz, making it thinner than the sample, which has a thickness of 1 mm. The high EMI shielding performance that has been achieved can be explained by this.[26]

The 10% CNTs-PP composite's unique structural and morphological characteristics, which result in the lowest electrical resistance, are responsible for its superior EMI shielding ability in the particular frequency range (4–9 GHz).[28]

Due to the high-speed impact during the ball milling process, conductive fillers (CNTs) are tightly coated on the surface of the HPP particles, which also prevents dust from contaminating the air.[44]

Because of their low weight, small diameter, high aspect ratio, high electrical conductivity, easy percolation, and good mechanical strength, carbon nanotubes (CNTs) have additional advantages over

conventional carbon fillers. When composited to polymers, high EMI SE could be easily achieved at a relatively low CNT percentage filling. The MWCNT/PMMA (in situ) EMI SE with 4.76 weight percent MWCNT shows EMI SE across a broad frequency range (leveled high value of 28.76 dB at 7.10 GHz and maximum SE of 32.06 dB at 2.18 GHz).[50]

An effective way to fabricate the above mentioned/desired EMI SE materials is using composite/nanocomposite shields. Functional and versatile EMI SE shields are essential, as is their cost-effective manufacturing at the required size and shape. Even less effective are polymeric matrices, unless they are conductive. A workable strategy for EMI SE is the harmonic use of conducting but nonmagnetic materials (like carbon) and magnetic alloys. There are two types of EMI shielding materials: structural and functional. When it comes to EM wave shielding, an NF-enforced composite outperforms its microfibre (MF) equivalent for polymeric matrices that are mainly transparent to radiation.[12]

LITERATURE SURVEY

S.No	Materials	Synthesis	Frequency	Thickness	Year	Reference
1.	Carbon Nanotube-Reinforced Polymer Composite	Melt blending, Ball milling	50 MHz – 8GHz	1 mm, 2.5mm	2020	[50]
2.	Polypropylene/Carbon Nanotubes (PP/CNT)	Melt blending	50 MHz – 8GHz	0.4 mm, 1 mm, 2.8mm	2020	[26]
3.	Polypropylene/Carbon Black (PP/CB)	Blending with CNT/CB powder	X-band (8.2 – 12.4 GHz)	2.5 mm	2022	[25]
4.	Graphene Nanoplatelets in Polypropylene	High-shear stress processing	5G frequency region	3 nm	2022	[29]
5.	Polyaniline/Graphite Composites	In situ emulsion processing	8.2 to 12.4 GHz	2 mm	2009	[36]
6.	Functionalized Lignin Nanoparticles with Mxene	Hydrothermal synthesis	8.2 to 12.4 GHz	600 nm	2024	[53]

7.	Multiwall Carbon Nanotubes/Polyp ropylene (MWCNT/PP)	Hydrothermal method, Ultrasonication	Ku-band: 18 to 27 GHz	45 μ m	2020	[50]
8.	Conducting Polymeric Composites	Oxidative Emulsion Polymerization	8.2 GHz	30 μ m	2020	[37]
9.	Structured Carbon from Waste Polymer	Carbonization of waste polymers	1-6 GHz	1.9mm	2024	[51]

A range of materials, such as carbon nanotubes (CNTs), graphene, polyaniline, lignin nanoparticles, and waste polymers, are utilized to enhance the electromagnetic interference (EMI) shielding properties of composites. CNTs and graphene are valued for their excellent electrical conductivity, while polyaniline serves as an effective conducting polymer. Functionalized lignin nanoparticles offer a sustainable option, and waste polymers contribute to the development of structured carbon materials with good shielding capabilities.

The synthesis methods for these materials vary widely, employing techniques like melt blending, ball milling, and hydrothermal synthesis to tailor their properties for specific shielding needs. For instance, polyaniline composites are created through oxidative emulsion polymerization, and waste polymers are carbonized to produce effective structured carbon materials. This diversity reflects ongoing research aimed at optimizing EMI shielding solutions.

These materials operate across a broad frequency range, from 50 MHz to 12.4 GHz, with some effective in the Ku-band and 5G frequency range, making them suitable for applications like microwave absorption and protection against harmful electromagnetic radiation. References for each material provide further insights into their properties and potential applications, highlighting the dynamic advancements in the field of EMI shielding.

CONCLUSION

Electromagnetic interference (EMI) shielding has emerged as an indispensable solution in modern society, where the exponential growth of electronic devices and wireless communication technologies has heightened the challenges of electromagnetic pollution. The disruption caused by EMI not only affects the functionality and performance of electronic devices but also poses significant health risks and contributes to environmental concerns. To mitigate these issues, the development of efficient, reliable, and sustainable EMI shielding materials has become a priority.

Traditional shielding materials, such as metals, offer high conductivity and effective EMI attenuation but are hindered by drawbacks, including their weight, susceptibility to corrosion, high cost, and rigidity. These limitations have driven the exploration of polymer-based composites as a superior alternative. Among these, polypropylene (PP) stands out due to its lightweight nature, excellent processability, recyclability, and resistance to chemical and environmental degradation. However, as a non-conductive material, pure PP lacks the inherent properties required for EMI shielding. This limitation is effectively addressed by the integration of conductive fillers like carbon nanotubes (CNTs), graphene, and metallic particles. These

fillers enable the creation of composite materials that exhibit high electromagnetic shielding effectiveness (EMI SE), along with enhanced mechanical and thermal properties.

The primary mechanisms for EMI shielding—reflection, absorption, and multiple internal reflections—underscore the importance of material conductivity, dielectric properties, and structural morphology. Conductive polymer composites (CPCs), particularly those based on PP reinforced with CNTs or steel fibers, leverage these mechanisms to achieve superior performance. Notably, PP composites have demonstrated the ability to balance lightweight and mechanical robustness with high shielding efficiency, making them ideal for applications in electronics, automotive, and aerospace industries.

Furthermore, advances in nanotechnology have significantly improved the effectiveness of these materials. The incorporation of hybrid fillers, such as MXene and graphene, has resulted in composites that can shield a wide frequency spectrum with minimal material thickness, achieving EMI SE values that rival or surpass traditional metals. These developments not only address the limitations of conventional materials but also align with the growing demand for environmentally sustainable and economically viable solutions.

In conclusion, the integration of polypropylene with advanced fillers represents a transformative approach to EMI shielding, combining the benefits of polymers with the superior electrical properties of conductive nanomaterials. The resulting composites are not only effective in shielding electromagnetic waves but also fulfill the requirements for flexibility, durability, and processability across diverse industries. As research continues to refine the synthesis methods and optimize material properties, the future holds immense potential for the widespread adoption of PP-based composites as a sustainable and efficient solution to the challenges posed by EMI. This progress will play a pivotal role in safeguarding both electronic systems and human well-being in an increasingly interconnected world.

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